Maxime Wabartha

maxime.wabartha@mail.mcgill.ca,
maxwab.github.io

EDUCATION

PhD in Computer Science, McGill University (Canada) 2019 - to date
Interpretability and reinforcement learning, advised by Prof. Joelle Pineau

MSc in Applied Mathematics, Université Paris-Saclay (France) 2017 - 2018
Master "Mathématiques, Vision, Apprentissage" (MVA). Topic: *Learning diverse neural networks for improved exploration in deep reinforcement learning*

BEng, MEng, École Centrale de Lille (France) 2010 - 2017

EXPERIENCE

Research intern, RBC Borealis (Canada) Sep. 2024 - Apr. 2025
Topic: *Off-policy evaluation in large action spaces* [1]

Visiting researcher, FAIR, Meta (Canada) Sep. 2022 - Sep. 2023
Topic: *Interpretability and deep reinforcement learning* [4]

Research intern, Polytechnique Montréal (Canada) May 2017 - Sep. 2017
Topic: *Semantic segmentation of the spinal cord* [9]

Business intelligence analyst, Shopwings (Australia) Jun. 2016 - Sep. 2016
Startup. Developing data analysis tools, project manager

Junior financial auditor, Ernst&Young (France) Sep 2015 - Mar. 2016
Financial audit of industrial french companies

Internal vice-president, Centrale Lille Projets (France) Apr. 2014 - Mar. 2015
Student-led consulting company (100k€ turn-over).
In charge of HR, project manager for 5 projects (∼15k€)

References available upon request

PUBLICATIONS & PREPRINTS

[1] **Wabartha, M.**, Wilson, K, Evans, D., Sharifi-Noghabi, H. & Sylvain, T.(2025). *Investigating Action Embeddings for More Efficient Off-Policy Evaluation.* RecSys workshop on Causality, Counterfactuals & Sequential Decision-Making.

[2] **Wabartha, M.** & Pineau, J. (2025). *Object-Centric Concept Representation and Use in RL Agents.* Under revision.

[3] Danesh, M., **Wabartha, M.**, Pineau, J. & Lin, H. C. (2025). *Mitigating Distribution Shifts: Uncertainty-aware Offline-to-online Reinforcement Learning.* Multidisciplinary Conference on Reinforcement Learning and Decision Making.

[4] **Wabartha, M.** & Pineau, J. (2024). *Piecewise Linear Parametrization of Policies for Interpretable Deep Reinforcement Learning.* International Conference on Learning Representations.

[5] **Wabartha, M.**, Durand, A., Francois-Lavet, V., & Pineau, J. (2020). *Handling Black Swan Events in Deep Learning with Diversely Extrapolated Neural Networks.* International Joint Conference on Artificial Intelligence, 2140-2147.

[6] Mangeat, G., Ouellette, R., **Wabartha, M.**, De Leener, B., Plattén, M., Danylaité Karrenbauer, V., ... & Granberg, T. (2020). *Machine Learning and Multiparametric Brain MRI to Differentiate Hereditary Diffuse Leukodystrophy with Spheroids from Multiple Sclerosis.* Journal of Neuroimaging.

[7] **Wabartha, M.**, Durand, A., François-Lavet, V., & Pineau, J. (2019). *Handling Black Swan Events in Deep Learning with Diversely Extrapolated Neural Networks.* NeurIPS Workshop on Safety and Robustness in Decision Making.

[8] **Wabartha, M.**, Durand, A., François-Lavet, V., & Pineau, J. (2018). *Sampling diverse neural networks for exploration in reinforcement learning.* NeurIPS Workshop on Bayesian Deep Learning.

[9] Zaimi, A.*, **Wabartha, M.***, Herman, V., Antonsanti, P. L., Perone, C. S., & Cohen-Adad, J. (2018). *AxonDeepSeg: automatic axon and myelin segmentation [...] using convolutional neural networks.* Nature Scientific reports, 8(1), 1-11.

* denotes an equal contribution.

SELECTED TALKS

Object-Centric Concept Representation and Use in RL Agents [2]
- Lab meeting, RBC Borealis — Mar. 2025
- Invited talk during 2-days lab visit, DtAK (Harvard University) — Nov. 2024

Improving the transparency of predictive models in the era of neural networks
- PhD proposal exam, McGill — Sep. 2023

Piecewise-linear parametrization of policies: towards interpretable deep reinforcement learning [4]
- Invited talk, DtAK (Harvard University) — Apr. 2024
- Research seminar, FAIR (Meta Montreal) — Sep. 2023
- Invited talk, EQUAL lab (McGill University) — Aug. 2023

Handling Black Swan Events in Deep Learning with Diversely Extrapolated Neural Networks [5]
- IJCAI '20, online — Jan. 2021

Using diverse ensembles for out-of-distribution detection [7]
- Invited talk, NeuroPoly lab (Canada) — Jun. 2019

SKILLS

Programming: Python, Pytorch, Jax, R

Software/OS: Git, Unix, Slurm, Streamlit , LaTeX, Matlab

AWARDS

FRQNT scholarship, doctoral program — 2021-2024
Fond de Recherche du Québec - Nature et Technologies
Competitive provincial scholarship, 25% acceptance

SERVICE

Reviewer: Reproducibility Challenge ('19, '20, '21 (Outstanding reviewer), '22), Montreal AI Symposium ('20), ECML ('22), ACM Computing Surveys ('24) ICLR ('25).

Volunteer helping with the organization of the RLDM conference in Montreal ('20)

LANGUAGES

French (native), English (fluent), Italian (conversational), German (conversational)

EXTRA-CURRICULAR

Practice of competitive badminton, 10 years

Camping